CS3570 Introduction to Multimedia Technology

Homework #1

Due: 11:59pm, 04/02/2026

1. Photo Enhancement (40%)

You are required to enhance the image "scene_dark.jpg" by following these steps.

(a) Convert the image from **RGB color space to YCbCr** . (5%)
(b) Apply **Gamma Transform** to the Y channel using an appropriate gamma value. (10%)
(c) Apply **Histogram Equalization** to the original Y channel. (10%)
(d) Convert the modified images (both Gamma Transform and Histogram Equalization) back from **YCbCr to RGB**, and display the enhanced images along with the histogram of the Y channel. (5%)
(e) Compare the original and enhanced images (both Gamma Transform and Histogram Equalization) along with their histograms. Discuss your observations regarding the differences and improvements. (10%)

Original Image

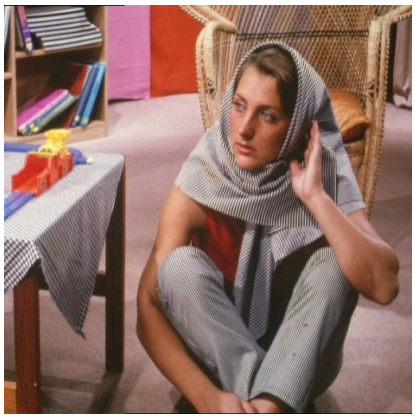
Enhanced Image

2. DCT Image Compression (60%)

Transform the image “Barbara.jpg“ from spatial domain to frequency domain with DCT for compression and quantize the frequency values with different quality factor. Then, reconstruct the compressed image using inverse DCT.
Please follow the steps below:

(a) Implement functions to convert images between RGB and **YCbCr** color spaces, including **4:2:0** chrominance subsampling and upsampling. (10%)
(b) Implement the **2D Discrete Cosine Transform (DCT) and Inverse Discrete Cosine Transform (IDCT)** functions for processing 8x8 pixel blocks. (10%)
(c) Implement a **quantization** function that scales the provided quantization tables based on a given quality factor and applies it to the DCT coefficients. (10%)

(d) Finish the complete compression process. Divide the image into 8x8 blocks, apply your functions from (a), (b), and (c), and then reconstruct the image back to the RGB color space. (10%)
(e) Calculate the **Peak Signal-to-Noise Ratio (PSNR)** and compare the original image with the reconstructed images across different quality factors. (10%)
(f) Brief report: Describe how you implemented the discrete cosine transform. (4%) Why should we convert image from RGB to YCbCr before compressing? (3%) Explain how can the quality factor affect the compression result. (3%)



Original Image

Compressed Image

Reminder

- You are **not allowed to use any function that directly generates the result** for each step, except for those provided by us.
- Your code must display and output your results to enable us to verify its correctness.
- Please follow the instructions in the Jupyter Notebook and complete the parts marked as "TODO."
- If you encounter any problems or have questions, please post them on eeclass.
- Rename your Jupyter Notebook file to Homework_01_xxxxxxxxx_ooo.ipynb, where xxxxxxxxx is your ID and ooo is your name. Ensure you upload it to eeclass before the deadline.
- Homework should be submitted before the announced due time. Scores of late homework will be reduced by 20% per day.